

PART A [25 marks]

This part consists of 3 questions. Answer **all** questions.

1. Evaluate the following limits.

(a) $\lim_{x \rightarrow -1} \frac{x+1}{3x^2+8x+5}$ [3 marks]

(b) $\lim_{x \rightarrow 5} \frac{\sqrt{x+4}-3}{x-5}$ [3 marks]

(c) $\lim_{x \rightarrow +\infty} \frac{2x+1}{\sqrt{x^2-9}}$ [3 marks]

2. Differentiate with respect to x and simplify your answers.

(a) $y = \frac{e^{5x}}{1+e^{2x}}$ [3 marks]

(b) $y = \ln(x^3\sqrt{2x^2-1})$ [4 marks]

(c) $y = \frac{\cos x}{1-\cos x}$, hence, find $\frac{dy}{dx}$ when $x = \frac{\pi}{2}$. [4 marks]

3. The curve $y = x^2 - \frac{16}{x}$ has a stationary point P. Find the coordinate of point P.

[5 marks]

PART B [75 marks]

This part consists of 7 questions. Answer **all** questions.

1. Solve the following.

a) Given that $z_1 = 1 + 2i$ and $z_2 = 3 - 4i$. Express $\frac{1}{z_1} + \frac{1}{z_1 - 2z_2}$ in the form

$a + bi$, where a and b are real numbers.

[5 marks]

b) Given that $z = x + yi$, where x and y are real numbers such that

$$\left| \frac{z}{z + 2 - i} \right| = \frac{1}{2}. \text{ Show that } 3x^2 + 3y^2 - 4x + 2y - 5 = 0.$$

[5 marks]

2. For each of the following.

a) If $\frac{e^{2x} - e^{-2x}}{e^{2x} + 2e^{-2x}} = \frac{1}{4}$, find the exact value of x .

[5 marks]

b) $\left| \frac{3x + 11}{x - 3} \right| > 4$, find the solution set for the inequality.

[6 marks]

3. Given that $f(x) = \begin{cases} -3, & x \leq -1 \\ x - 2, & -1 < x \leq 4 \\ 2 - (x - 4)^2, & x > 4 \end{cases}$

a) Sketch the graph of $y = f(x)$

[3 marks]

b) State the domain and range of f .

[2 marks]

c) Determine whether f is a one-to-one function or not. Give reason for

your answer.

[2 marks]

4. Function f and g are defined as follows.

$$f(x) = \frac{x+2}{x-4} \quad \text{and} \quad g(x) = 3x-2$$

- a) If $(f \circ g)(x) = 3$, determine the value of x . [4 marks]
- b) Show that $f^{-1}(x) = \frac{2(2x+1)}{x-1}$. State the domain and range of f^{-1} . [4 marks]
- c) Find $f^2(x)$. [3 marks]

5. a) Given that $f(x) = \frac{x}{(x+1)(x-3)^2}$.

- i. Show that function f has two vertical asymptote and state its equation. [3 marks]
- ii. Find the limits at infinity. Deduce the equation of the horizontal asymptote. [3 marks]

b) A function f is defined by
$$f(x) = \begin{cases} \frac{x-1}{x+2}, & 0 \leq x < 2 \\ ax^2 - 1, & x \geq 2 \end{cases},$$

where $a \in \mathbb{R}$. Find the value of a such that $\lim_{x \rightarrow 2} f(x)$ exists. With this value of

a , determine whether f is continuous at $x = 2$.

[5 marks]

6. a) If $e^y = 3x - 4y$, find $\frac{dy}{dx}$ in terms of x and y . Hence, show that

$$(3x - 4y + 4)^2 \frac{d^2y}{dx^2} + 3(3x - 4y) \frac{dy}{dx} = 0$$

[5 marks]

- b) A curve has the parametric equations $x = \frac{1+t}{1-t}$ and $y = \frac{2}{t^2}$, where t is a parameter.

Show that

i. $\frac{dx}{dt} = \frac{2}{(1-t)^2}$, [2 marks]

ii. $\frac{d}{dt} \left(\frac{dy}{dx} \right) = \frac{2(3-t)(1-t)}{t^4}$, [5 marks]

iii. Find the value of $\frac{d^2y}{dx^2}$ at the point (3,8). [4 marks]

7. A hollow right circular cone has a base radius of 5 cm and a height of 10 cm. A cylinder is inscribed in the cone such that the axes of the cone and cylinder coincide.

- a) If the cylinder has a base radius of x cm and a height of h cm, show that the volume of the cylinder is $2\pi x^2(5-x)\text{cm}^3$.

[3 marks]

- b) Find the maximum volume of the cylinder.

[6 marks]

END OF QUESTION PAPER

PART A

1. (a) $\frac{1}{2}$ b) $\frac{1}{6}$ c) 2

$$2. \quad \text{(a)} \quad \frac{5e^{5x} + 3e^{7x}}{(1 + e^{2x})^2} \qquad \text{b)} \quad \frac{8x^2 - 3}{x(2x^2 - 1)}$$

c) $\frac{-\sin x}{(1-\cos x)^2}; \frac{dy}{dx} = -1$

3. $(-2, 12)$

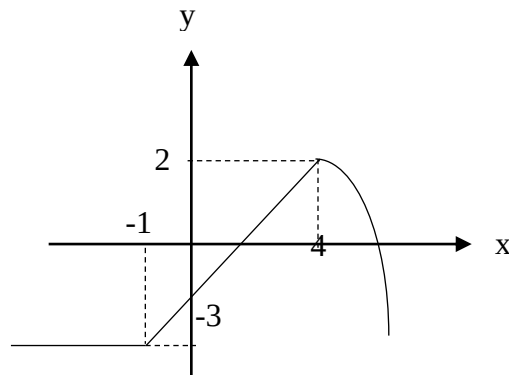
PART B

1. (a) $\frac{4}{25} - \frac{12}{25}i$ x

2. (a) $\frac{1}{4} \ln 2$

(b) $\left(\frac{1}{7}, 3\right) \cup (3, 23)$

3. (a)



b) $D_f = (-\infty, \infty); R_f = (-\infty, 2]$

c) By using horizontal line test (x-axis), intersect the graph at two points.

Not one-to-one function

4. a) $x = 3$
- b) $D_{f^{-1}} = (-\infty, 1) \cup (1, \infty); R_{f^{-1}} = (-\infty, 4) \cup (4, \infty)$
- c) $\frac{x-2}{6-x}$
5. (a) i. $x = -1$ or $x = 3$
- ii $y = 0$
- (b) $a = \frac{5}{16}$; continuous at $x = 2$
6. (a) $\frac{dy}{dx} = \frac{3}{3x-4y+4}$
- (b) iii. 5
7. (b) $\frac{1000\pi}{27} \text{ cm}^3$